Exercise Sheet 2 System Safety

Exercise 2.1: Safety Vocabulary *optional*

Define the following terms in your own words and give a concrete example from an autonomous driving context for each:

1. **Loss**
2. **Hazard**
3. **Risk**

1. **Loss:**
   A loss is an adverse outcome involving harm to people, property, or the environment that must be prevented.
   Example: Vehicle hits a pedestrian → injury or death.

2. **Hazard:**
   A hazard is a system state or condition that, under certain circumstances, can lead to a loss.
   Simple understanding: Dangerous situation that can cause a loss.
   Example: Vehicle is moving while a pedestrian is in its path. (Important: accident has not happened yet, but it can happen.)

3. **Risk:**
   Risk is the combination of the likelihood of a hazard occurring and the severity of the resulting loss.
   Simple understanding: How likely something bad is + how bad it would be.
   Example: Missing pedestrian detection
   Likelihood: medium
   Severity: very high
   → high risk

Loss → outcome (damage)
Hazard → situation (danger)
Risk → evaluation (likelihood + severity)

STPA: System-Theoretic Process Analysis

STPA Step 1: Define Purpose and Identify Losses, Hazards, and ODD

Exercise 2.2: ODD Specification

Based *only* on the system description in the course Roadmap (you have not yet seen the training data), define the **Operational Design Domain** (ODD) for the CARLA system. Cover at least the following dimensions: weather, lighting, camera condition, scene type, and vehicle speed.

For each dimension, state:

1. The **operating conditions** (where the system is considered valid).
2. The **non-operating conditions** (where it is not).
3. How an ODD violation could be **detected at runtime**.

Hint: You are defining the ODD *before* looking at the data. In Exercise Sheet 3, you will compare this ODD to the actual training data and (possibly) discover gaps.

1. **ODD**:
   The Operational Design Domain (ODD) is the **set of conditions under which the system is intended to operate safely**. Outside this domain, no safety guarantees apply. [simple exp: Where the system is allowed to work.]

2. **CARLA** = a simulated world to train and test self-driving systems.

3. **Dimensions**:

   1. Weather
   - Operating: clear, sunny, light clouds
   - **Non-operating:** heavy rain, fog, snow
   - **Detection of violation:**
   use camera visibility checks (blur, low contrast) or weather sensors

2. Lighting
   - **Operating**: daytime
   - **Non-operating**: night, low-light condition.
   - **Detection**:
   measure brightness of image (pixel intensity)

3. Camera condition
   - Operating: clean, fixed, forward-facing
   - Non-operating: dirty, occluded, misaligned

- Detection: detect blur, obstruction, or abnormal image patterns

4. Scene type

- Operating: urban environment (roads, intersections)
- Non-operating: highways, rural roads, unusual environments
- Detection:
scene classification model or map-based verification

5. Vehicle speed

- Operating: low speed (e.g., city/intersection driving)
- Non-operating: high-speed driving
- Detection:
check speed sensor values

Main idea:

- Inside ODD → system is expected to work
- Outside ODD → behavior is unreliable
- **System must detect ODD violations**

Exercise 2.3: Losses

STPA begins by identifying the **losses** that must be prevented - unacceptable outcomes involving harm to people, property, or the environment.

For each loss, identify its ID (L-1, L-2, ...), briefly describe it, and explain why it is unacceptable:

ID	Loss Description	Why Unacceptable
L-1	Loss of vehicle control leading to unsafe behavior	Can lead to accidents involving multiple participants.
L-2	Collision causing damage to vehicles or infrastructure	Financial damage and possible secondary risks
L-3	Injury or death of a pedestrian	Direct harm to human life
L-4	Violation of traffic laws	Legal consequences and unsafe system behavior

L-5	Injury to passengers inside the vehicle	The system must also ensure safety of its own occupants

Exercise 2.4: Hazards

A **hazard** is a system state that, combined with worst-case environmental conditions, can lead to a loss. Hazards are properties of the *system*, not of individual components.

1. Identify at least four system-level hazards (H-1, H-2, ...) for the CARLA system.
2. For each hazard, state which loss(es) from Exercise 2.3 it can lead to.
3. Rate each hazard's likelihood (low / medium / high) and its severity (low / medium / high).

Fill in the following table:

ID	**Hazard**	**Loss(es)**	**Likelihood**	**Severity**
H-1	Vehicle is moving while a pedestrian is in its path	L-3	Medium	High
H-2	Vehicle fails to detect another vehicle or obstacle in front	L-2, L-1	Medium	High
H-3	Vehicle continues driving despite unsafe or unknown environment (outside ODD)	L-1, L-2, L-3	High	High
H-4	Planner issues incorrect control action (eg: accelerates instead of braking)	L-1, L-2, L-5	Medium	High
H-5	Traffic light is detected as present, but its state (red/green) is unknown, leading to unsafe decisions at intersections	L-2, L-3, L-4	Medium	High

STPA Step 2: Model the Control Structure

Exercise 2.5: Control Structure Diagram

Draw the control structure for the CARLA system. Your diagram must include:

- All components listed in the system description (camera, perception models, distance oracle, planning module, actuators, human operator).
- **Control actions** (solid arrows): commands flowing from controllers to controlled processes (e.g., "brake command").
- **Feedback** (dashed arrows): information flowing back (e.g., "perception labels", "distance estimates", "vehicle speed").

Hint: There are at least two controllers in this system: the planning module and the human operator. The perception models and distance oracle should be considered *sensor/feedback*, not controllers themselves - they provide information to the controllers.

Hint: The traffic-light model only detects presence, not traffic light state. Consider what decisions the planner can (or cannot) safely make with this information.

1. **Control Structure**:
   A control structure represents the system as **controllers, controlled processes, control actions, and feedback loops**.

   It shows:
   - who makes decisions
   - what actions are taken
   - what information is used

2. **Diagram:**

STPA Step 3: Identify Unsafe Control Actions and Derive Safety Constraints

Exercise 2.6: Unsafe Control Actions (UCAs)

Using your control structure, identify **unsafe control actions** for at least **two different controllers**. For each UCA, also indicate which hazard(s) from Exercise 2.4 it could cause or lead to (reference them by ID, e.g., “H-1”, “H-2”).

For each controller, consider all four UCA types:

1. **Not provided** - a required action is missing.
2. **Provided unsafely** - the action is issued when it should not be.
3. **Wrong timing / order** - too early, too late, or out of sequence.
4. **Wrong duration** - applied too long or stopped too early.

Fill in the table below (add rows as needed):

ID	Controller	Control action	UCA type	Hazard(s)	Unsafe scenario
UCA-1	**Planning module**	Brake command	—---	H1, H2	Planner does not issue brake when a pedestrian or obstacle is in path
UCA-2	Planning module	Continue command	Provided unsafely	H-1, H-2, H-3	Planner continues driving despite unsafe environment or obstacle
UCA-3	Planning module	Brake command	Wrong timing	H1, H2	Brake is applied too late to avoid collision
UCA-4	Planning module	Brake command	Wrong duration	H1, H2	Brake is released too early before stopping

UCA-5	**Human operator**	Override command	—--	H-3, H-4	Human does not intervene when system behaves unsafely
UCA-6	Human operator	Override command	Provided unsafely	H-2, H-4	Human overrides correct system decision and causes unsafe action
UCA-7	Human operator	Override command	Wrong timing	H-1, H-2	Human intervenes too late to prevent accident
UCA-8	Human operator	Override command	Wrong duration	H-4	Human releases or applies control for incorrect duration

Exercise 2.7: Safety Constraints

For each UCA from Exercise 2.6, derive at least one **safety constraint** - a requirement that, if satisfied, prevents the UCA from occurring or limits its consequences.

Classify each constraint as either:

- **Model-level** - can be verified by testing or evaluating the corresponding perception model (this is what you will do in Sheets 4–9).
- **System-level** - requires a design measure beyond the model (e.g., redundancy, operator protocol, fallback behavior).

UCA	Safety constraint	Level	Verification
UCA-1	The planner must issue a braking command when a pedestrian or obstacle is within stopping distance	System-level	Scenario-based testing / simulation

UCA-2	The planner must not continue driving when the environment is unsafe or uncertain	System-level	ODD violation tests / safety logic checks
UCA-3	The system must ensure braking is initiated within a safe time threshold	System-level	Latency measurement / timing tests
UCA-4	The braking action must be maintained until the vehicle reaches a safe stop	System-level	Simulation tests with stopping distance
UCA-5	The system must alert the human operator when unsafe conditions occur	System-level	Human-in-the-loop testing
UCA-6	The system must prevent unsafe overrides or provide warnings to the operator	System-level	Override safety checks / interface testing

STPA Step 4: Identify Causal Loss Scenarios

Exercise 2.8: Causal Loss Scenarios

For each UCA from Exercise 2.6, identify **at least one causal scenario** - a specific mechanism or sequence of events that could cause the UCA to occur in practice.

For each scenario, describe:

1. **The causal factor:** What component failure, misunderstanding, or adverse condition triggers the UCA?
2. **Why it occurs:** What is the root cause or underlying reason?
3. **Connection to constraint:** Which safety constraint from Exercise 2.7 is designed to prevent or mitigate this scenario?

Hint: Consider the following causal mechanisms (from the lecture):

- **Incorrect feedback:** A sensor or perception model provides wrong information to the controller.
- **Flawed internal model:** The controller misinterprets correct information (e.g., low confidence treated as "no detection").
- **Timing delay:** Information arrives too late for the controller to respond safely.
- **Human factor:** The human operator fails to intervene due to attention lapse, delayed reaction, or mode confusion.

UCA	Causal scenario	Root cause	Related constraint
UCA-1	Pedestrian is present but not detected and the planner does not issue brake	Perception model misses pedestrian	Planner must issue braking command when obstacle is within stopping distance
UCA-2	System is outside ODD (eg: night or low light conditions), but planner continues driving	No ODD violation detection	Planner must not continue driving in unsafe or unknown conditions
UCA-3	Detection arrives late and the braking command issued too late	High latency in perception or processing	System must ensure braking within safe time threshold
UCA-4	Vehicle starts braking but stops too early and hits obstacle	Incorrect distance estimation or control logic	Braking must continue until safe stop is reached
UCA-5	System behaves unsafely but human does not intervene	Human inattention or lack of alert	System must alert operator in unsafe situations
UCA-6	Human reacts too late to prevent collision	Delayed perception or slow reaction time	System must provide timely alerts for intervention

Connecting STPA to the Rest of the Course

Exercise 2.9: Mapping Constraints to Evidence *optional*

Review the safety constraints you derived in Exercise 2.7 and the causal scenarios from Exercise 2.8. For each **model-level** constraint, identify which future exercise sheet will provide evidence (refer to the Course Roadmap). For each **system-level** constraint, briefly describe what kind of evidence or argument you would need - even though you cannot run an experiment for it in this course.